

Retrospective Confirmation for Identity-Consistent Streaming 3D Perception

Anonymous ICCV submission

Paper ID 0000

Abstract

not deliver at the same output budget, from a training-free module that transfers across pipelines unchanged.

Streaming 3D perception systems emit objects whose identities and class labels flicker: the same physical object is re-detected as a new instance, or changes class, from one frame to the next. We present a training-free retrospective confirmation module that improves this identity and semantic consistency without retraining or modifying the perception model. Detections are associated into tracks and emitted only after N observations, the standard confirmation test. At confirmation the module retrospectively emits the withheld prefix — a step standard online trackers omit — thereby recovering complete object trajectories. The module changes no box coordinate, score, or label, runs on frozen detector output, and transfers across pipelines at a single operating point ($N=3$). Because it also reduces how much a system emits, and most metrics respond to output volume alone, we validate it against matched controls rather than an unfiltered baseline: a score threshold calibrated to the identical detection count, and a per-sequence top- K selection matched to the identical identity count. On nuScenes with a frozen detector, the module improves association accuracy against both controls in both sensor- and world-frame association, with bootstrap confidence intervals excluding zero, and raises class-aware tracking accuracy at an equal detection budget (AMOTA 0.094 $\rightarrow$ 0.134 and 0.314 $\rightarrow$ 0.386; point estimates, consistent in sign across both association frames). On ScanNet200, an indoor open-vocabulary instance-segmentation pipeline sharing no detector, modality, or codebase with the outdoor one, the same module reduces class-label switching by 26.9% against an identity-matched control (309 of 312 scenes) with segmentation AP unchanged, while a random selection of equal size reproduces none of the reduction. The gain is in consistency specifically, and it is paid for: at an equal detection budget the module lowers mAP, detection recall, and aggregate HOTA, and retrospective emission costs $N-1$ frames of latency, without which the association gain holds in the sensor frame but not the world frame. What survives the controls is the contribution: identity and semantic consistency that score thresholding does

1. Introduction

Streaming 3D perception systems — detectors and semantic mappers that run on a live sensor feed [8–10, 13] — are consumed downstream as *persistent objects*: a map, a scene graph, a planner’s object memory. What those consumers need is that one physical object stay one object with one label. What streaming systems actually produce flickers. The prevailing emission rule is a per-frame confidence threshold: every detection above a score cut-off enters the stream immediately, independently of what was emitted a frame earlier. Nothing carries an object’s history forward, so a re-detected object arrives as a new identity, and an unstable open-vocabulary label changes class along the way. Per-frame detection metrics do not see any of this, and so do not penalise it. Nor does the usual demonstration of a temporal fix settle the matter: a component that suppresses unstable output is conventionally compared against an unfiltered baseline, a comparison that cannot separate what the component *selects* from how much it *removes*. The property is thus both unmeasured by the dominant metrics and unattributed by the dominant comparison.

We present a *retrospective confirmation* module that improves identity and semantic consistency in such a stream, and requires no retraining and no change to the perception model. Detections are linked into tracks by a class-agnostic associator; a track becomes *confirmed* only after N observations, and at that moment the module emits not only the current detection but the track’s withheld prefix, so a confirmed object’s trajectory is complete from its first observation. Nothing about a box’s coordinates, score, or label is modified. The retrospective step, which ordinary online trackers [2, 23, 24] do not perform, is what makes the consistency gain hold in both of the association regimes we test, and it is what the module contributes beyond the confirmation test it builds on. The module runs at a single operating point ($N=3$) with no per-dataset or per-metric tuning.

The attribution problem stated above applies to our mod-

ule as much as to any other: confirmation also changes the *amount* of output, and every volume-sensitive metric moves for that reason alone. We therefore evaluate it under a protocol built to settle exactly that question — non-temporal *matched controls*, which hold the output budget fixed so that only the selection rule varies. A *detection-budget-matched* control thresholds the same frozen detector output so that it emits exactly as many detections as the module; an *identity-budget-matched* control keeps, per sequence, the highest-scoring K tracks, where K is the number of identities the module emits. Both controls share the detector, the association step, and the evaluation code with the module, and differ from it only in selecting by score instead of by temporal persistence.

Under these controls, on the nuScenes validation set (150 scenes, a frozen CenterPoint [26] detector, both sensor-frame and ego-motion-compensated world-frame association, confirmed tracks emitted retrospectively), the module improves identity consistency and pays for it in per-frame detection quality. It improves association accuracy (HOTA’s AssA component [16]) over both matched controls in both association frames, with bootstrap confidence intervals excluding zero. The class-aware tracking accuracy AMOTA [4], which does not admit the same per-scene resampling and is reported as a point estimate and a pre-registered secondary endpoint, rises consistently in sign in both frames under detection-budget matching; under identity matching we do not read it as an independent effect, for a reason given in Sec. 5.2. At the same output budget it *reduces* per-frame detection quality, a cost we quantify in Sec. 5.1. We present both sides as the finding: the benefit of temporal confirmation is identity consistency, and it is bought rather than added.

Because the module touches only emission timing, it transfers to any streaming pipeline that re-observes objects. We apply the identical module, unchanged and at the same N , to an indoor open-vocabulary instance-segmentation pipeline that shares no detector, modality, or codebase with the outdoor one, and evaluate it under the same control design. On ScanNet200 (312 validation scenes, 3D instance-mask proposals labeled per frame by a 2D open-vocabulary detector), the module reduces the number of frames in which an instance’s class label changes by 26.9% relative to an identity-budget-matched control of equal size, while leaving instance-segmentation AP essentially unchanged. The temporal system has fewer label switches in 309 of 312 scenes and the paired confidence interval excludes zero by a wide margin, whereas selecting the same number of instances at random reproduces none of the reduction — so the gain is specific to temporal selection rather than to emitting fewer identities.

Contributions.

- A training-free retrospective confirmation module.

An operator combining the standard confirmation test with retrospective emission of the confirmed prefix, applied to frozen detector output at a single operating point, with no retraining, no architectural change, and no modification of box coordinates, scores, or labels.

- **The retrospective emission policy, isolated.** We show the retrospective step is what completes the module: emitting causally instead narrows *where* the gain holds — it survives in the sensor frame but reverses in the world frame — and retrospection’s price is $N-1$ frames of latency (1.0 s at 2 Hz for $N=3$).
- **A matched-control evaluation protocol.** Two non-temporal controls — detection-budget-matched and identity-budget-matched — with paired per-sequence statistics, which separate what temporal selection contributes from what merely emitting less contributes. The protocol is not specific to confirmation and applies unchanged to any component that alters output volume.
- **Cross-domain validation under that protocol.** Outdoors, the module improves association accuracy under both controls in both association frames, and class-aware tracking accuracy at a matched detection budget in both frames, at a detection cost we quantify in Sec. 5.1 rather than leave implicit. Indoors, the identical module reduces class-label switching on an unrelated open-vocabulary instance-segmentation pipeline, with segmentation quality unchanged.

2. Related Work

Streaming perception. Streaming perception was formalised by Li et al. [13], who observed that a detector evaluated offline is scored on a world that has already moved on, and introduced streaming AP (sAP): predictions are compared against the state of the world at the moment they are *emitted*, so latency and accuracy are folded into one number. Follow-up work optimises detectors directly for this regime [25], and the 3D analogue processes LiDAR sectors as they arrive rather than waiting for a full sweep [9]. This line establishes the premise we build on — that a perception system on a live stream must be judged as a stream — but its correction is about *when* a prediction is scored. sAP is still a per-frame average precision: two systems that emit identical boxes at identical latency but differ entirely in identity continuity and label stability receive the same sAP. Our focus is orthogonal and complementary: we hold latency and box geometry fixed and ask what the per-frame family cannot see *within* a frame-aligned comparison.

Multi-object tracking. Online tracking-by-detection descends from SORT [2], which pairs a Kalman filter with greedy Hungarian IoU association, and DeepSORT [24], which adds an appearance embedding to survive occlusion. Both use a *track-confirmation* rule: a tentative track is not

reported until it has been matched in N consecutive frames. The confirmation test is inherited from this literature; what has not been examined is what it contributes once the volume it removes is held fixed. In 3D, AB3DMOT [23] showed that the same approach applied to 3D boxes is a strong baseline, and CenterPoint [26] tracks by predicting velocity and associating centre points greedily across frames. Throughout this literature, confirmation is treated as a tuned nuisance parameter and compared against an unfiltered tracker, so its measured benefit includes whatever follows from emitting less output. We instead compare it against non-temporal selection rules granted the same output budget, and report what changes on the detection metrics the 3D perception field reports as well as on the tracking metrics.

Open-vocabulary 3D perception. OpenScene [17] distills 2D vision-language features into 3D points for zero-shot querying without 3D annotation, and LERF [11] embeds language into a radiance field. OpenMask3D [21] moves to instances, computing per-mask CLIP features over class-agnostic 3D proposals, and Open-YOLO 3D [3] replaces the expensive per-mask feature extraction with cheap 2D open-vocabulary box predictions projected onto 3D masks. All four are *offline*: they consume a completed reconstruction. ConceptFusion [10] makes the setting online, fusing open-set multimodal features into a map frame by frame, and ConceptGraphs [8] builds an open-vocabulary 3D scene graph incrementally, associating and merging per-frame detections into persistent object nodes for planning. Once a system is incremental, *which* per-frame detection becomes which persistent object is a first-class question — yet evaluation remains per-scene AP or mIoU on the final map, which is insensitive to how often a persistent object’s identity or class label changes along the way. Our indoor experiment measures exactly that property, on a pipeline of this family, under a matched control.

Evaluation metrics. CLEAR MOTA [1] scores a tracker by counting FP, FN, and identity switches, but is dominated by detection errors; IDF1 [18] rebalances toward identity by matching whole trajectories one-to-one and reporting an identity F1. HOTA [16] was introduced precisely because these conflate two capabilities, and factorises the score into a detection accuracy DetA and an association accuracy AssA, whose geometric mean is HOTA. This factorisation is what makes our result legible: association quality and detection quality can be read separately over an *identical* box set, which is precisely the distinction a temporal selection rule acts on. In 3D, nuScenes AMOTA [4] — following AB3DMOT’s integral formulation [23] — averages MOTA over recall thresholds and is class-aware, so it responds to class labels that the class-agnostic family ignores; we report both. Video consistency metrics (STQ [22], VPQ [12], TETA [14]) and open-vocabulary

tracking benchmarks (TAO [7], OVTrack [15]) extend the family but likewise require dense GT tube identities over a fixed taxonomy. We use these metrics where they are computable rather than proposing a new one; our contribution is the controlled experiment and the protocol that reads it.

Controlled evaluation of pipeline components. Isolating what a single component contributes requires holding the rest of the system fixed, and for components that change how much output a system produces, holding the output budget fixed as well; otherwise the comparison rewards volume rather than the mechanism. Budget-matched comparison is standard practice when scoring detectors at a fixed number of proposals, but it is not usually applied to temporal components, whose effect on output volume is a side effect rather than a parameter. The controls in this paper apply it in both of the currencies a temporal component spends — detections and identities — and read the resulting trade-off directly.

3. Method

Throughout this paper, *module* denotes a reusable post-processing operator composed of three existing parts — a class-agnostic associator, the standard confirmation test for track initialization, and an emission policy — applied to frozen detector output (Fig. 1). The module is deliberately minimal: no part of it changes a box’s coordinates, score, or label, and it decides only *which* detections enter the output stream and under *which* persistent identity. The confirmation test descends directly from tracking-by-detection practice [2, 23, 24]; the retrospective-emission completion and the controlled evidence for what the composition does (Sec. 4) are what we add.

3.1. Streaming pipeline

Per-frame detections (a 3D box with a score and a class label) are linked across frames into tracks by a class-agnostic nearest-centroid associator with a 2.0 m bird’s-eye-view center-distance threshold and a gap tolerance of 5 frames; no score threshold is applied before association. Association is performed either in the *sensor frame* (raw detector coordinates) or in the *world frame* (ego-motion-compensated global coordinates); both conditions are evaluated throughout, since ego-motion compensation changes how far the same physical object moves between frames and hence the difficulty of association. The associator is identical in every arm of every experiment. It is deliberately lightweight rather than state of the art, because the quantity of interest is the difference a selection rule makes and not the highest attainable tracking score; absolute tracking numbers reported here are correspondingly lower than those of tuned nuScenes trackers and are not comparable to them.

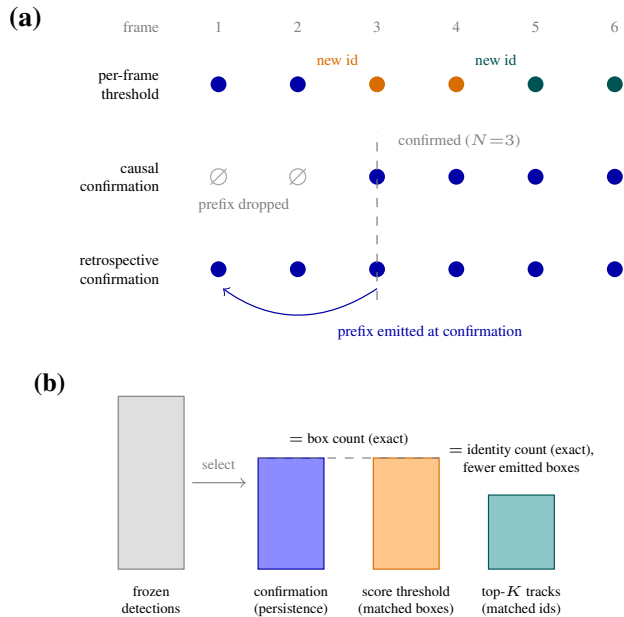


Figure 1. The module and the matched controls. **(a)** Emission policies for one re-observed object over six frames. A per-frame score threshold emits immediately but carries no history forward, so re-detections arrive as new identities (color changes). Confirmation-based initialization emits a track only once it has N observations ($N=3$ shown): the *causal* policy drops the pre-confirmation prefix, while the *retrospective* policy emits the withheld prefix at confirmation time, completing the trajectory at a latency of up to $N-1$ frames. **(b)** Every arm selects from the same frozen detection set. The detection-budget control is a score threshold calibrated to the module’s exact box count; the identity-budget control keeps the module’s track count per sequence, which necessarily emits fewer boxes because score-ranked tracks are shorter (Sec. 4.2).

3.2. Confirmation-based track initialization

A newly created track is *tentative*. It becomes *confirmed* once it has accumulated N observations; only confirmed tracks contribute detections to the output stream, and tracks that die before confirmation are never emitted. We use a single operating point, $N=3$, throughout the paper — the conventional setting in tracking-by-detection systems — rather than tuning N per dataset or per metric. The module is therefore a pure temporal *selection*: the emitted detections are a subset of the baseline detector’s output, chosen by track persistence instead of by score.

3.3. Emission policy

Confirmation leaves a policy choice for the first $N-1$ observations of a track that does get confirmed (Fig. 1a). A *causal* (online) policy emits nothing until confirmation and drops the pre-confirmation detections; an *offline retrospective* policy emits the withheld prefix at confirmation time, so a confirmed object’s trajectory is complete from its first

observation. The retrospective policy is only available to consumers that tolerate output latency: a detection may appear up to $N-1$ frames after it was observed — at the 2 Hz annotated-frame rate of our outdoor benchmark, 2 frames or 1.0 s for $N=3$. We adopt the retrospective policy for all main results and state this latency alongside every table; the causal policy is evaluated as an ablation (Sec. 5.3), since the two policies emit different numbers of boxes by construction.

4. Evaluation Protocol

The protocol is designed around one confound: confirmation reduces output volume, and the detection-oriented metrics this field reports are strongly coupled to how much a system emits, which makes a consistency gain hard to tell apart from the effect of suppression alone. Every claim in this paper is therefore made against a *matched control* — a non-temporal selection rule granted the same output budget — rather than against the unfiltered baseline.

4.1. Datasets and pipelines

Outdoor. The nuScenes [5] validation split, all 150 scenes at the 2 Hz annotated-frame rate. Detections come from a pretrained CenterPoint [26] detector whose per-frame output is computed once from 10-sweep LiDAR input and *frozen*; every arm of every experiment replays the same stored detections (718,786 boxes over the split; mAP 0.5613, NDS 0.6480), so any metric difference between arms is attributable to the selection rule, not to detector variance.

Indoor. The ScanNet200 [19] validation split, all 312 scenes, in a streaming instance-segmentation pipeline: class-agnostic 3D instance-mask proposals from Mask3D [20] are labeled per frame by a 2D open-vocabulary detector [6], so each persistent instance receives a stream of per-frame class labels. The proposal set is likewise frozen across arms. This pipeline shares no detector, modality, or codebase beyond the temporal module with the outdoor one, which is the point: it tests whether the same confirmation rule transfers.

4.2. Matched controls

Two controls bracket the two budgets a temporal module alters (Fig. 1b).

Detection-budget-matched control. A score threshold is applied to the frozen baseline detections and calibrated so that the control emits *exactly* as many boxes as the temporal module — 257,259 boxes in the sensor frame (threshold 0.2214) and 517,566 in the world frame (threshold 0.1272), both with zero count error. The match is exact rather than approximate because the count is a step function of the threshold whose breakpoints are the observed scores: the matched threshold is the K -th largest observed score, with

ties resolved upward so the count never exceeds K . This control answers: does temporal selection beat the detector’s own confidence ranking at equal detection volume?

Identity-budget-matched control. Per sequence, the control keeps the K highest-scoring tracks (ranked by mean detection score), where K is the number of identities the temporal module emits in that sequence. The track counts match exactly (41,207 tracks in the sensor frame; 68,222 in the world frame). This control answers: does temporal selection beat score-based selection at equal *identity* volume? Five additional arms select the same K tracks per sequence *at random* (five seeds), scored on association accuracy, separating score-based selection from selection per se. One budget cannot be matched at a time without unmatching the other: at equal track count the control emits fewer boxes than the temporal module (138,053 vs. 257,259 sensor; 412,203 vs. 517,566 world), because score-ranked tracks are shorter than persistence-selected ones. We report this asymmetry explicitly and interpret the two controls jointly rather than reading either alone.

Indoor control. The identity-budget design transfers directly: the control keeps the K highest-scoring instances per scene, where K matches the temporal module’s emitted-instance count (10,413 instances in total; the control realizes 10,410). Three additional arms select K instances *at random* per scene (three seeds), to separate the effect of score-based selection from that of selection per se.

Shared machinery. Baseline, temporal module, and every control share the frozen detections, the associator, and the evaluation code; they differ only in the selection rule.

4.3. Metrics

Outdoor. Official nuScenes mAP and NDS on the emitted box set; class-agnostic HOTA, AssA, DetA, detection recall (DetRe), and IDF1 via TrackEval [16, 18], with similarity $\text{sim} = \text{clip}(1 - d_{xy}/2.0, 0, 1)$ matching the associator’s 2.0 m threshold; and official class-aware AMOTA [4] over the 7 nuScenes tracking classes. The class-agnostic and class-aware families are reported side by side because they respond to different properties of the stream.

Indoor. The pre-registered primary endpoint is the per-scene count of *label switches*: frames in which a persistent instance’s emitted class label differs from its label in the previous frame *in which it was emitted*. Counting against the last emission rather than the last frame is what makes the count comparable across arms, since the arms emit on different frames. The secondary endpoint is instance-segmentation AP on the final reconstruction, with the pre-registered expectation of no material change.

4.4. Statistics

All three main comparisons — detection-budget-matched, identity-budget-matched, and indoor — were pre-

registered: before each run we fixed the arms, the primary endpoint, and the criterion for a negative result. For the outdoor comparisons that criterion was that the association gain hold in *both* association frames with scene-bootstrap intervals excluding zero, and that the dataset-level metrics (mAP, AMOTA) move consistently across the two association frames, whichever direction they move in; indoors, that the per-scene label-switch count fall. The pre-registration documents will be released with the code. The causal-emission ablation (Sec. 5.3) was *not* pre-registered; it precedes those registrations and is what motivated them. All comparisons between the temporal module and a control are paired per scene. We report the per-scene Wilcoxon signed-rank test with the rank-biserial correlation as effect size, and a 95% bootstrap confidence interval on the difference of the aggregate metric obtained by resampling scenes (10,000 resamples, fixed seed 20260718). A difference is described as an improvement only when its interval excludes zero; intervals that span zero are reported as no detectable difference, whatever their sign.

Intervals are available for the class-agnostic metrics, which are defined per scene and aggregate linearly over scenes. AMOTA is not, and neither are the nuScenes devkit metrics mAP and NDS: AMOTA integrates MOTA over recall thresholds estimated on the whole split, and mAP and NDS are likewise computed over the whole split, so a per-scene resample does not yield the same estimator. We report all three as point estimates without intervals. We therefore do not claim interval support for AMOTA anywhere; the evidence we offer for it is that its sign is stable across the four detection-budget-matched comparisons (retrospective and causal emission, each in both association frames). Under identity matching AMOTA is reported as a pre-registered secondary endpoint only (Sec. 5.2).

5. Experiments

We report three outdoor comparisons on nuScenes val and one indoor comparison on ScanNet200 val. Each is a comparison between the confirmation-based module and a non-temporal control granted the same output budget, over frozen detections shared by both arms. Unless stated otherwise, outdoor results use the offline retrospective emission policy, whose latency cost is $N-1$ frames (1.0 s at the 2 Hz annotated-frame rate).

5.1. Detection-budget-matched comparison

Table 1 compares confirmation-based initialization against a score threshold calibrated to the identical box count. The comparison is therefore free of the volume confound: both arms emit 257,259 boxes in the sensor frame and 517,566 in the world frame, drawn from the same frozen detector output.

Frame	Arm	Boxes	mAP	NDS	HOTA	AssA	DetA	DetRe	IDF1	AMOTA
Sensor	Baseline (unfiltered)	718,786	0.5613	0.6480	0.1793	0.2007	0.1620	0.6690	0.1112	0.0858
	Control (threshold)	257,259	0.5504	0.6450	0.2735	0.2119	0.3570	0.6160	0.2082	0.0942
	Confirmation ($N=3$)	257,259	0.3009	0.5229	0.2462	0.2744	0.2232	0.4287	0.2054	0.1338
World	Baseline (unfiltered)	718,786	0.5613	0.6480	0.2837	0.5014	0.1613	0.6663	0.2715	0.2896
	Control (threshold)	517,566	0.5598	0.6479	0.3305	0.5146	0.2133	0.6550	0.3391	0.3137
	Confirmation ($N=3$)	517,566	0.4921	0.6200	0.3224	0.5293	0.1973	0.6141	0.3288	0.3864

Table 1. **Detection-budget-matched comparison** on nuScenes val (150 scenes, frozen CenterPoint detections, retrospective emission). The control is the same detection set thresholded at a score calibrated to emit *exactly* the same number of boxes as the confirmation module (0.2214 sensor, 0.1272 world; zero count error). At equal detection budget, confirmation improves association accuracy (AssA) and class-aware tracking accuracy (AMOTA) in both frames, and loses detection-oriented accuracy (mAP, DetA, DetRe) and aggregate HOTA. Bold marks the better of the two budget-matched arms, and is omitted where the difference has a confidence interval spanning zero (sensor-frame IDF1). The unfiltered baseline is shown for reference only and is not budget-matched.

Association accuracy improves. AssA rises from 0.2119 to 0.2744 in the sensor frame and from 0.5146 to 0.5293 in the world frame. Both differences have scene-bootstrap 95% confidence intervals excluding zero ($[+0.0471, +0.0785]$ and $[+0.0098, +0.0203]$). Per scene, the sensor-frame gain is $+0.0460$ on average — a mean of per-scene differences, a different estimator from the 0.0625 difference of the aggregates above — and holds in 113 of 150 scenes (Wilcoxon $p=2.6 \times 10^{-15}$, rank-biserial 0.74).

Class-aware tracking accuracy rises. AMOTA goes from 0.0942 to 0.1338 (sensor) and from 0.3137 to 0.3864 (world), an increase of $+0.040$ and $+0.073$ absolute. As noted in Sec. 4.4, AMOTA does not admit the per-scene resampling used for the other metrics, so this is a point estimate; what we rely on is that it moves the same way in both association frames here and under causal emission below (Sec. 5.3).

Detection-oriented metrics fall. At the same budget, mAP drops $0.5504 \rightarrow 0.3009$ (sensor) and $0.5598 \rightarrow 0.4921$ (world), DetA drops $0.3570 \rightarrow 0.2232$ ($[-0.1408, -0.1267]$) and $0.2133 \rightarrow 0.1973$ ($[-0.0183, -0.0138]$), and detection recall drops $0.6160 \rightarrow 0.4287$ ($[-0.2087, -0.1665]$) and $0.6550 \rightarrow 0.6141$ ($[-0.0477, -0.0350]$); mAP is a dataset-level metric and is reported as a point estimate. Because HOTA is the geometric mean of DetA and AssA, the DetA loss dominates and aggregate HOTA falls as well ($[-0.0323, -0.0231]$ sensor; $[-0.0100, -0.0063]$ world). The trade-off is explicit: the module buys association quality with detection quality rather than adding it on top.

IDF1 shows no detectable aggregate sensor-frame difference. The bootstrap interval on the sensor-frame IDF1 difference spans zero ($[-0.0108, +0.0037]$), so by the criterion of Sec. 4.4 we report no detectable difference in aggregate IDF1. The per-scene paired test does not indicate parity, however: it leans against the confirmation module ($p=0.007$; confirmation ahead in 65/150 scenes). We therefore read sensor-frame IDF1 as at best unchanged and

Arm	Boxes	HOTA	AssA	DetA	IDF1	AMOTA
<i>Sensor frame</i>						
Control (top- K)	138,053	0.2959	0.2607	0.3383	0.2542	0.1004
Control (rand.)	—	—	.2595–.2765	—	—	—
Confirmation	257,259	0.2462	0.2744	0.2232	0.2054	0.1338
<i>World frame</i>						
Control (top- K)	412,203	0.3630	0.5074	0.2610	0.3969	0.2911
Control (rand.)	—	—	.4705–.4776	—	—	—
Confirmation	517,566	0.3224	0.5293	0.1973	0.3288	0.3864

Table 2. **Identity-budget-matched comparison.** The control keeps, per scene, the K highest-scoring tracks, with K set to the number of identities the confirmation module emits (41,207 sensor, 68,222 world; exact match in both frames). The random- K rows (“rand.”) give the range over five seeds of the same selection made at random, scored on the claim-bearing metric (AssA) only. AMOTA is a pre-registered secondary endpoint. Matching identities leaves the box budgets unequal (the control emits fewer boxes), which handicaps the control on AMOTA, so its AMOTA column is not read as an independent effect of confirmation. In the sensor frame the random- K range contains the confirmation module’s AssA.

possibly slightly lower, and claim no gain. In the world frame IDF1 is slightly *lower* for the confirmation module ($[-0.0135, -0.0070]$).

5.2. Identity-budget-matched comparison

Matching the number of emitted identities rather than the number of boxes (Tab. 2) tests whether the association gain merely reflects having fewer identities to confuse. Against the score-ranked control it does not. AssA remains higher for the confirmation module in both frames ($0.2607 \rightarrow 0.2744$ sensor, $0.5074 \rightarrow 0.5293$ world), with intervals excluding zero ($[+0.0071, +0.0216]$ and $[+0.0177, +0.0267]$) and per-scene wins in 107/150 and 116/150 scenes ($p=3.9 \times 10^{-15}$ and 2.9×10^{-16}). Selecting the same number of tracks *at random* rather than by score (Sec. 4.2) behaves differently in the two frames. In

Arm	mAP	HOTA	AssA	AMOTA
<i>Sensor frame</i>				
Control (threshold)	0.5287	0.3044	0.2147	0.0997
Confirmation, causal	0.2047	0.2484	0.3088	0.1290
<i>World frame</i>				
Control (threshold)	0.5562	0.3864	0.5308	0.3417
Confirmation, causal	0.4480	0.3451	0.5184	0.4050

Table 3. **Causal emission, detection-budget-matched** (168,193 boxes sensor at threshold 0.3320, 361,426 world at 0.1673; both arms exact). Dropping the pre-confirmation prefix instead of emitting it retrospectively preserves the AMOTA gain in both frames, but it reverses the world-frame association result: AssA is lower than the control, with a confidence interval entirely below zero. For this ablation we report confidence intervals for AssA only, the metric the comparison turns on; the remaining columns are point estimates.

the world frame it does not reproduce the result: the five seeded arms reach at most 0.4776, below the confirmation module’s 0.5293. In the sensor frame the random arms are *not* separated from the confirmation module — they span 0.2595–0.2765, a range that contains its 0.2744 — so the sensor-frame identity-matched result rests on the score-ranked control alone (Tab. 2). AMOTA, the pre-registered secondary endpoint, is higher for confirmation than for the top- K control (0.1004 $\rightarrow$ 0.1338 sensor, 0.2911 $\rightarrow$ 0.3864 world), but we do not read this as an independent effect: the control emits fewer boxes, which limits the recall over which AMOTA is integrated, so part of the difference is the box-volume imbalance rather than confirmation.

Budget asymmetry. The two controls cannot be satisfied simultaneously. At equal identity count the control emits fewer boxes than the confirmation module (138,053 vs. 257,259 sensor; 412,203 vs. 517,566 world), because score-ranked tracks are shorter than persistence-selected ones. The imbalance cuts both ways: it handicaps the control on AMOTA, which is therefore reported but not interpreted here, and the control’s advantage on DetA and HOTA in this table is partly a budget artifact and should be read from Tab. 1, where boxes are matched exactly. IDF1 likewise favours the control here (Tab. 2), which the control’s shorter tracks help; we read it together with Tab. 1, where it shows no detectable aggregate sensor-frame difference. The two tables are meant to be read together: the association gain over the score-based controls survives both matchings, and the detection loss is established under the matching that isolates it.

5.3. Emission policy

The results above emit a confirmed track’s withheld prefix retrospectively, which costs 1.0 s of latency. Table 3 reports the causal alternative, in which the prefix is permanently dropped, again against a count-matched score

threshold. The result splits by association frame. The AMOTA gain survives in both frames (0.0997 $\rightarrow$ 0.1290 sensor, 0.3417 $\rightarrow$ 0.4050 world), and in the sensor frame the association gain is not merely preserved but larger than under retrospective emission (0.2147 $\rightarrow$ 0.3088; interval [+0.0722, +0.1162] against [+0.0471, +0.0785] retrospectively, 112 of 150 scenes). In the world frame, however, causal emission *loses* association accuracy relative to the control (0.5308 $\rightarrow$ 0.5184), with a bootstrap interval entirely below zero ([−0.0195, −0.0049]; paired $p=0.036$, rank-biserial −0.20, with confirmation ahead in only 59 of 150 scenes). The detection cost is larger throughout than under retrospective emission (mAP 0.5287 $\rightarrow$ 0.2047 sensor).

The latency-free policy is therefore not uniformly worse — it is worse in exactly one place, and that place is where association was already easy. Ego-motion compensation removes most of the apparent motion between frames, so the world-frame control associates well on its own (AssA 0.5308 against 0.2147 in the sensor frame); there the detections lost with the discarded prefix outweigh what confirmation cleans up. Where association is hard, the opposite holds. We offer that reading as an interpretation of the pattern rather than something we measured directly, but the operational consequence stands on the measurements alone: a system that cannot absorb $N-1$ frames of latency can still use confirmation, and should expect the association gain only if it associates in sensor coordinates.

5.4. Confirmation-window sensitivity

All results so far fix $N=3$. Table 4 varies it over $\{2, 3, 4, 5\}$, giving each N its own exactly matched control, and asks what the choice is worth. $N=1$ is excluded: it confirms every track on its first observation, reproduces the unfiltered baseline detection-for-detection, and its matched control is therefore the same box set, so every difference is zero by construction. We use it only as an implementation check.

The association gain does not depend on the operating point. It is positive at every N in both association frames, and all eight intervals exclude zero. It also grows monotonically with N — in the sensor frame close to linearly (+0.0296, +0.0625, +0.0933, +0.1221), in the world frame with visible deceleration (+0.0103, +0.0148, +0.0181, +0.0199). The class-aware metric behaves differently in the two frames. In the sensor frame AMOTA keeps rising through the range (+0.0478 $\rightarrow$ +0.0529 from $N=4$ to $N=5$); in the world frame it turns over, peaking at $N=4$ (+0.0736) and falling at $N=5$ (+0.0682). The detection cost, meanwhile, rises without turning over at any N we measured, and the latency rises linearly by construction.

No single N in this range is best on every metric, and $N=3$ is not the best on any of them. We report this rather than adjust the operating point: $N=3$ is the conventional default we fixed before running these experiments and used

Frame	N	Delay	Boxes	Δ AssA	95% CI	Δ AMOTA	Δ DetRe
Sensor	2	1	375,982	+0.0296	[+0.0217, +0.0383]	+0.0176	-0.1210
	3	2	257,259	+0.0625	[+0.0471, +0.0785]	+0.0396	-0.1873
	4	3	203,019	+0.0933	[+0.0720, +0.1149]	+0.0478	-0.2217
	5	4	171,395	+0.1221	[+0.0960, +0.1480]	+0.0529	-0.2408
World	2	1	594,547	+0.0103	[+0.0068, +0.0140]	+0.0614	-0.0278
	3	2	517,566	+0.0148	[+0.0098, +0.0203]	+0.0726	-0.0409
	4	3	457,121	+0.0181	[+0.0120, +0.0250]	+0.0736	-0.0492
	5	4	409,463	+0.0199	[+0.0130, +0.0276]	+0.0682	-0.0556

Table 4. **Confirmation-window sensitivity.** Each row is confirmation at that N with retrospective emission, minus *its own* detection-budget-matched threshold control, matched exactly to the box count in the Boxes column. Delay is $N-1$ frames (0.5 s each at 2 Hz). The $N=3$ rows are the comparison of Tab. 1, reproduced here rather than recomputed. Δ AMOTA is a whole-split point estimate; the DetRe intervals are omitted for space and exclude zero at every N . This sweep is a sensitivity analysis and was not pre-registered.

unchanged in every result in this paper and in both domains, and re-selecting it from the table above would convert an inherited constant into a value tuned on the test metrics — the practice the matched controls exist to guard against. What the sweep establishes is the claim we do make: the consistency gain is a property of confirmation over the whole range tested, not an artifact of one setting, and larger windows buy more of it at proportionally more detection cost and more latency. Beyond demonstrating that the conclusion is not specific to a single operating point, the monotonic increase in AssA with larger confirmation windows is consistent with the proposed confirmation mechanism: longer confirmation accumulates more evidence, producing progressively stronger identity consistency after exact detection-budget matching. The widening gap is driven by the confirmed arm rather than by a failing control, whose own AssA also rises as the matched budget tightens (0.2080 $\rightarrow$ 0.2146 sensor, 0.5061 $\rightarrow$ 0.5260 world, from $N=2$ to $N=5$).

5.5. Indoor: semantic stability

The indoor experiment asks whether the same rule improves *semantic* stability in a pipeline that shares no detector, modality, or codebase with the outdoor one. Table 5 reports the pre-registered primary endpoint. Against the identity-matched control, confirmation reduces label switches from 23,303 to 17,045, a 26.9% reduction at an essentially equal instance budget (10,413 vs. 10,410). Per scene the reduction is 18.0 switches at the median (20.1 on average, 95% CI on the mean [18.7, 21.5]) and holds in 309 of 312 scenes (Wilcoxon $p=1.5 \times 10^{-52}$, rank-biserial 0.99).

The three random- K arms (23,275, 23,277, 23,288 switches) are indistinguishable from the score-ranked control and far from the confirmation module, so the reduction is specific to temporal selection rather than to emitting fewer instances by any rule.

The pre-registered secondary endpoint behaves as expected: instance segmentation AP is 0.1954 against 0.1956 for the unfiltered baseline and 0.1926 for the control, i.e. no

Arm	Inst.	Switches	AP	AP ₅₀
Baseline (unfiltered)	—	—	0.1956	—
Control (top- K)	10,410	23,303	0.1926	0.2504
Control (random K)	—	23,275–23,288	—	—
Confirmation ($N=3$)	10,413	17,045	0.1954	0.2532

Table 5. **Indoor comparison** on ScanNet200 val (312 scenes, frozen Mask3D proposals with per-frame open-vocabulary labels). The primary endpoint is the per-scene count of label switches (lower is better); the control keeps the K highest-scoring instances per scene with K matched to the confirmation module’s emitted-instance count. The random- K row spans three seeds (23,275, 23,277, 23,288) and is scored on the primary endpoint only. Bold marks the primary endpoint, the only pre-registered endpoint in this table; the AP columns are secondary and are not tested. The unfiltered baseline is the frozen validity anchor, for which only AP was recorded; it is not an arm of the comparison, hence its empty cells.

material change. Unlike the outdoor setting, the indoor gain therefore does not come at a measurable detection-quality cost — the two pipelines differ in whether the withheld detections carry segmentation mass, and we report the difference in effect without claiming the outdoor trade-off is avoidable in general.

6. Discussion

What the module improves, and what the controls add. Compared against an unfiltered baseline, the module appears to improve nearly every tracking metric. Compared against a non-temporal rule with the same output budget, part of that apparent improvement turns out to be the effect of emitting less, and what remains is a sharper claim: association accuracy and class-aware tracking accuracy improve, detection-oriented accuracy degrades, and identity F1 shows no detectable aggregate gain at a matched detection budget (and is lower in the world frame) and is lower at a matched identity budget, where the control’s shorter tracks favour it. The surviving gain is the one worth having,

because it is the one a score threshold cannot buy. The controls are what let us say that; they are the evidence standard behind the method’s claim rather than a separate finding, and the general lesson — a component that spends output volume must be compared at fixed volume — is available to anyone evaluating a similar module.

Association and detection respond to opposite things. What a confirmation rule removes is fixed by construction: tracks observed fewer than N times. The falling detection recall shows directly that some of those withheld detections matched ground truth, which is the cost side of the trade-off. The gain side admits an interpretation we can offer but did not measure separately — that the identities surviving confirmation are the ones an associator handles most reliably. What the measurements do establish is that the two halves move in opposite directions over an identical box set, which is why we report DetA and AssA rather than HOTA alone: the aggregate hides a trade-off it is averaging over.

The two domains agree in direction, not in cost. Outdoors the identity gain is paid for in detection quality; indoors the analogous semantic-stability gain leaves AP unchanged. The shared finding is that in both pipelines temporal selection improves a property the per-frame metric does not measure — identity consistency outdoors, label stability indoors — across two systems with nothing in common but the rule. The differing cost is a property of the pipelines, and we do not generalise from the indoor result that the outdoor cost can be avoided.

Practical reading. Within the settings we measured — this detector’s error profile on nuScenes, and the indoor pipeline on ScanNet200 — the case our evidence favours is a system whose output is consumed as persistent objects: a map, a scene graph, a planner’s object memory. A system scored on per-frame detection quality should not adopt confirmation-based initialization. Latency decides not whether to adopt it but what to expect: with $N-1$ frames of slack the association gain holds in both association frames, and without that slack it holds in the sensor frame but not the world frame (Sec. 5.3). Those conditionals are the recommendation the controls support, and they are narrower than the one an uncontrolled comparison would license.

7. Limitations

One tracking-GT dataset. The full tracking evaluation (HOTA family, AMOTA) is nuScenes only. ScanNet200 has no instance-tracking ground truth, so the indoor endpoint is label switching rather than an identity metric, and the two domains are not measured with the same instrument. Multi-dataset tracking validation (Waymo, Argoverse) remains future work.

One operating point. We fix $N=3$ and do not tune it per dataset or metric. The value is inherited: it is the conven-

tional default in tracking-by-detection systems [2, 24], not a setting we chose by inspecting our own results, and the same value is used in both domains. This avoids selection on the test metric. The outdoor sweep of Sec. 5.4 shows the association gain holds across $N \in \{2, 3, 4, 5\}$, so the claim is not specific to $N=3$; it also shows $N=3$ is not the best setting on any single metric, and that the class-aware metric turns over inside the range in the world frame (but not the sensor frame) while the class-agnostic one does not in either. We did not sweep N indoors, and we do not know where either curve ends beyond $N=5$.

Budget matching is not simultaneous. Matching detections leaves identity counts unequal and vice versa (Sec. 5.2). We report both matchings and their directions of bias rather than a single comparison that is matched in both currencies, which does not exist for a rule that changes track lengths.

Latency. All main results use retrospective emission and are therefore not achievable online at zero latency. The causal variant is measured (Sec. 5.3) and preserves the association gain only in the sensor frame.

Frozen detector. Every arm replays one stored detection set, which is what makes the comparison clean, but it also means we characterise the rule’s effect for a single detector’s error profile. A detector with fewer short-lived false positives has less for confirmation to remove.

8. Future Work

Two of the limitations above point in the same direction. Because the causal variant forfeits the world-frame association gain and the retrospective one buys it with $N-1$ frames of latency, the interesting design space is between them: a hybrid policy that emits provisionally and revises, or one that varies the confirmation depth with association confidence or scene dynamics, would trade latency against identity consistency continuously rather than at the two endpoints we measure. Sec. 5.4 maps that trade-off over N outdoors; doing the same over emission policy, and indoors, is the natural next measurement, and one the controls of Sec. 4 support without modification. The same holds for the association step: the associator here is deliberately lightweight, and substituting a stronger one, or a different detector or dataset, changes what is being measured but not how to measure it.

9. Conclusion

We presented a training-free retrospective confirmation module for streaming 3D perception: detections are linked into tracks, a track enters the output stream only after N observations, and its withheld prefix is then emitted retrospectively. It requires no retraining, leaves box coordinates, scores, and labels untouched, and runs at one operat-

ing point across two pipelines that share nothing else. On nuScenes with a frozen detector it improves association accuracy and class-aware tracking accuracy in both association frames; on ScanNet200 it reduces class-label switching by 26.9% with segmentation AP unchanged. Both results are established against non-temporal controls granted the module’s own output budget — in detections and in identities — and against a random-selection control indoors, so what they credit is temporal selection rather than the smaller volume it emits. The gain is bought rather than added: at an equal detection budget the module trades away per-frame detection quality (Sec. 5.1), and the retrospective step costs $N-1$ frames of latency (Sec. 5.3). On the identity-consistency and label-stability axes — and on those axes alone — a system whose output is consumed as persistent objects is, on this evidence, better served by temporal confirmation than by the score threshold it would otherwise spend the same budget on. In practice that points at consumers for which identity persistence and label stability are worth more than minimum-latency detection — long-horizon tracking, mapping, planning, and interaction systems — and away from any system scored on per-frame detection quality. Any future component that changes how much a system emits can be held to the same standard with the same controls.

References

- [1] Keni Bernardin and Rainer Stiefelhausen. Evaluating multiple object tracking performance: The CLEAR MOT metrics. *EURASIP Journal on Image and Video Processing*, 2008.
- [2] Alex Bewley, Zongyuan Ge, Lionel Ott, Fabio Ramos, and Ben Upcroft. Simple online and realtime tracking. In *ICIP*, 2016.
- [3] Mohamed El Amine Boudjoghra et al. Open-YOLO 3d: Towards fast and accurate open-vocabulary 3d instance segmentation. In *NeurIPS*, 2024.
- [4] Holger Caesar et al. nuScenes tracking benchmark (AMOTA/AMOTP), 2019.
- [5] Holger Caesar et al. nuScenes: A multimodal dataset for autonomous driving. In *CVPR*, 2020.
- [6] Tianheng Cheng et al. YOLO-World: Real-time open-vocabulary object detection. In *CVPR*, 2024.
- [7] Achal Dave et al. TAO: A large-scale benchmark for tracking any object. In *ECCV*, 2020.
- [8] Qiao Gu, Alihusein Kuwajerwala, Sacha Morin, Krishna Murthy Jatavallabhula, et al. ConceptGraphs: Open-vocabulary 3d scene graphs for perception and planning. In *ICRA*, 2024.
- [9] Wei Han et al. Streaming object detection for 3-d point clouds. In *ECCV*, 2020.
- [10] Krishna Murthy Jatavallabhula et al. ConceptFusion: Open-set multimodal 3d mapping. In *RSS*, 2023.
- [11] Justin Kerr et al. LERF: Language embedded radiance fields. In *ICCV*, 2023.
- [12] Dahun Kim et al. Video panoptic segmentation. In *CVPR*, 2020.
- [13] Mengtian Li, Yu-Xiong Wang, and Deva Ramanan. Towards streaming perception. In *ECCV*, 2020.
- [14] Siyuan Li et al. Tracking every thing in the wild. In *ECCV*, 2022.
- [15] Siyuan Li et al. OVTrack: Open-vocabulary multiple object tracking. In *CVPR*, 2023.
- [16] Jonathon Luiten et al. HOTA: A higher order metric for evaluating multi-object tracking. *IJCV*, 2021.
- [17] Songyou Peng et al. OpenScene: 3d scene understanding with open vocabularies. In *CVPR*, 2023.
- [18] Ergys Ristani et al. Performance measures and a data set for multi-target, multi-camera tracking. In *ECCV Workshops*, 2016.
- [19] David Rozenberszki, Or Litany, and Angela Dai. Language-grounded indoor 3d semantic segmentation in the wild (ScanNet200). In *ECCV*, 2022.
- [20] Jonas Schult et al. Mask3D: Mask transformer for 3d semantic instance segmentation. In *ICRA*, 2023.
- [21] Ayca Takmaz et al. OpenMask3D: Open-vocabulary 3d instance segmentation. In *NeurIPS*, 2023.
- [22] Mark Weber et al. STEP: Segmenting and tracking every pixel. In *NeurIPS Datasets and Benchmarks*, 2021.
- [23] Xinshuo Weng, Jianren Wang, David Held, and Kris Kitani. 3D multi-object tracking: A baseline and new evaluation metrics. In *IROS*, 2020.
- [24] Nicolai Wojke, Alex Bewley, and Dietrich Paulus. Simple online and realtime tracking with a deep association metric. In *ICIP*, 2017.
- [25] Jinrong Yang et al. Real-time object detection for streaming perception. In *CVPR*, 2022.
- [26] Tianwei Yin, Xingyi Zhou, and Philipp Krähenbühl. Center-based 3d object detection and tracking. In *CVPR*, 2021.